$$g(1)S(n)=\sum_{i=1}^{n}g(i)S(\lfloor\frac{n}{i}\rfloor)-\sum_{i=2}^{n}g(i)S(\lfloor\frac{n}{i}\rfloor)=\sum_{i=1}^{n}(f*g)(i)-\sum_{i=2}^{n}g(i)S(\lfloor\frac{n}{i}\rfloor)$$